Laboratory Project: Intelligent Position Tracking Control of a 2-DOF Robot Manipulator Using Fuzzy Logic Control

Control Systems Laboratory

Abstract

The objective of this laboratory project is to design and simulate an intelligent controller for a two-degree-of-freedom (2-DOF) robot manipulator. A discrete-time mathematical model of the manipulator is developed using Euler discretization. A fuzzy logic controller is employed to achieve accurate position tracking despite the nonlinear and coupled dynamics of the robot. The complete simulation is implemented in C language using Dev-C++, and the generated data are stored in a text file for visualization and analysis using MATLAB. The obtained results are used to evaluate the tracking performance, transient response, and steady-state accuracy of the intelligent controller.

Keywords: 2-DOF Robot Manipulator, Intelligent Control, Fuzzy Logic Control, Position Tracking, Dev-C++, MATLAB.

1 Introduction

Robot manipulators are widely used in industrial automation, manufacturing, assembly, and precision positioning applications. Due to the highly nonlinear and coupled nature of robot dynamics, conventional linear controllers may not provide satisfactory performance under varying operating conditions.

Intelligent control techniques such as fuzzy logic controllers have become attractive alternatives because they can handle nonlinearities and uncertainties without requiring an exact mathematical model of the system.

The objective of this laboratory project is to design and implement a fuzzy logic position controller for a 2-DOF robot manipulator. The entire simulation must be developed using the C programming language in Dev-C++, and the generated results must be stored in a text file and analyzed using MATLAB.

The specific objectives are:

- Develop the dynamic model of a 2-DOF robot manipulator.
- Derive its discrete-time representation.
- Design a fuzzy logic controller.

- Implement the simulation in Dev-C++.
- Store simulation results in a text file.
- Analyze tracking performance using MATLAB.

2 Discrete Mathematical Model of the 2-DOF Robot Manipulator

2.1 Continuous-Time Dynamic Model

The dynamics of a two-link rigid robot manipulator are described by

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau \tag{1}$$

where

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \tag{2}$$

and

$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} \tag{3}$$

represent the joint positions and applied torques, respectively.

2.2 Inertia Matrix

The inertia matrix is

$$M(q) = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \tag{4}$$

with

$$M_{11} = a + 2b\cos(\theta_2) \tag{5}$$

$$M_{12} = M_{21} = d + b\cos(\theta_2) \tag{6}$$

$$M_{22} = d \tag{7}$$

where

$$a = I_1 + I_2 + m_1 l_{c1}^2 + m_2(l_1^2 + l_{c2}^2) \tag{8}$$

$$b = m_2 l_1 l_{c2} \tag{9}$$

$$d = I_2 + m_2 l_{c2}^2 \tag{10}$$

2.3 Coriolis and Centrifugal Matrix

$$C(q, \dot{q}) = \begin{bmatrix} -b\sin(\theta_2)\dot{\theta}_2 & -b\sin(\theta_2)(\dot{\theta}_1 + \dot{\theta}_2) \\ b\sin(\theta_2)\dot{\theta}_1 & 0 \end{bmatrix} \tag{11}$$

2.4 Gravity Vector

$$G(q) = \begin{bmatrix} (m_1 l_{c1} + m_2 l_1) g\cos(\theta_1) + m_2 l_{c2} g\cos(\theta_1 + \theta_2) \\ m_2 l_{c2} g\cos(\theta_1 + \theta_2) \end{bmatrix} \tag{12}$$

2.5 State Variables

Define the state vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \dot{\theta}_1 \\ \theta_2 \\ \dot{\theta}_2 \end{bmatrix} \tag{13}$$

2.6 State Equations

The state equations are

$$\dot{x}_1 = x_2 \tag{14}$$

$$\dot{x}_2 = \ddot{\theta}_1 \tag{15}$$

$$\dot{x}_3 = x_4 \tag{16}$$

$$\dot{x}_4 = \ddot{\theta}_2 \tag{17}$$

where

$$\ddot{q} = M^{-1}(q)\left[\tau - C(q, \dot{q})\dot{q} - G(q)\right] \tag{18}$$

2.7 Euler Discretization

Using a sampling period

$$T_s = 0.001 \ s \tag{19}$$

the discrete-time model becomes

$$x_1(k+1) = x_1(k) + T_s x_2(k) \tag{20}$$

$$x_2(k+1) = x_2(k) + T_s \ddot{\theta}_1(k) \tag{21}$$

$$x_3(k+1) = x_3(k) + T_s x_4(k) \tag{22}$$

$$x_4(k+1) = x_4(k) + T_s \ddot{\theta}_2(k) \tag{23}$$

2.8 Desired Trajectories

For the simulation, the desired joint trajectories are chosen as

$$\theta_{1d}(t) = 0.5 \sin(2\pi t) \tag{24}$$

$$\theta_{2d}(t) = 0.5 \cos(2\pi t) \tag{25}$$

for

$$0 \leq t \leq 10 \ s \tag{26}$$

2.9 Fuzzy Logic Controller Design

For each robot joint, define the tracking error

$$e_i = \theta_{id} - \theta_i \tag{27}$$

and its derivative

$$\dot{e}_i = \frac{e_i(k) - e_i(k-1)}{T_s} \tag{28}$$

The controller inputs are

$$[e_i, \dot{e}_i] \tag{29}$$

and the output is the control torque

$$\tau_i \tag{30}$$

Students must design a Mamdani fuzzy controller using the following linguistic variables:

- NB (Negative Big)
- NM (Negative Medium)
- NS (Negative Small)
- Z (Zero)
- PS (Positive Small)
- PM (Positive Medium)
- PB (Positive Big)

The controller must include:

- Membership functions

- Rule base
- Fuzzification stage
- Defuzzification stage

3 Simulation Requirements

The simulation program shall be implemented in Dev-C++ using the C language.

The program must:

1. Initialize robot parameters.
2. Generate desired trajectories.
3. Compute tracking errors.
4. Execute the fuzzy controller.
5. Calculate joint torques.
6. Update robot states.
7. Store simulation results in a file named `data.txt`.

The following variables must be stored:

`t    theta1_d    theta1    theta2_d    theta2`

using

```
fprintf(fp,
"%lf %lf %lf %lf %lf\n",
t,
theta1_d,
theta1,
theta2_d,
theta2);
```

4 Results and Discussion

Students must use MATLAB to plot the desired and actual joint positions.

The following figures are required:

- Joint 1 Position Tracking
- Joint 2 Position Tracking

The discussion must include:

- Rise Time

- Settling Time
- Maximum Overshoot
- Steady-State Error
- Tracking Accuracy
- Controller Robustness

Students must comment on the effectiveness of fuzzy logic control for nonlinear robot systems.

5 Conclusion

A complete intelligent control system for a 2-DOF robot manipulator has been designed and implemented. The robot dynamics were modeled and discretized using Euler approximation. A fuzzy logic controller was employed to achieve accurate trajectory tracking. Simulation results obtained from Dev-C++ and MATLAB demonstrated the capability of intelligent control techniques to improve tracking performance in nonlinear robotic systems.

References

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